

EKF-Based Sensorless IPM Synchronous Motor Drive for Flux-Weakening Applications

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Abstract - The paper deals with the experimental realisation of a sensorless interior permanent magnet (IPM) synchronous motor drive. The motor has been specifically designed to get a wide flux-weakening region, to minimise the drive power rating in spite of a wide speed range. Position and angular speed of the rotor are obtained through an Extended Kalman Filter (EKF). The estimation algorithm requires neither the knowledge of the mechanical parameters, nor the initial rotor position. In the paper, particular emphasis is placed on control algorithms, that are complicated by the motor anisotropy and that have been specifically studied to enhance the overall system performance.

I. INTRODUCTION

Several industrial and automotive applications call for drives operating above the motor base speed, in the so-called flux-weakening (FW) region. For permanent magnet synchronous motors (PMSM), this reflects in a particular rotor design, with internal permanent magnet (IPM) placement. Buried magnets create a rotor anisotropy, that in turn yields several advantages, as extended flux weakening range, high torque-to-ampere and torque-to-volume ratios [1,2]. A simple manufacture should also assure a cost-effective end product. Compared to conventional induction motors or surface mounted PMSMs, IPM motors require a more complex drive control. A bright idea for effective operations in the flux-weakening region is reported in [3], in which an external voltage loop is used to calibrate the FW action. The same principle has been exploited in [4], where an IPM motor drive has been applied to an electrical scooter, to minimise the drive power rating in spite of a wide speed range. The demand of inexpensive and reliable drives now pushes applied research towards the elimination of mechanical sensors [5-8]. The next challenge for IPM drives is thus their reliable and robust sensorless control. Since an IPM motor behaves like a salient-pole machine, stator inductances are functions of the rotor position. Former speed and position observers were based on this feature. The inductance can be obtained by monitoring the stator voltages and currents, and the inductance variations can be used to estimate the rotor position and speed [9, 10].

A promising rule is played by extended Kalman filter (EKF) techniques, for their ability to perform state estimation for non-linear systems, with inherent robustness against parameters variation. The theoretical aspects on EKF technique have been deeply investigated in the literature [11-14]. There are also application examples [14,15], in which well-known pitfalls, as the starting from unknown rotor position and the filter matrices tuning have been successfully fixed. Anyway, it comes out that the residual weakness of sensorless control resides in the low-speed range, in which noise and inverter non-ideality still play a harmful rule.

This paper presents the extension of the EKF-based sensorless technique to IPM synchronous motor drives, for FW-oriented applications. The implicit idea is to exploit the excellent properties of EKF at high speed, that is, far from standstill critical neighbouring.

The task is not trivial, since the rotor anisotropy complicates the mathematical model. In Sect. II, a short review of IPM motor structure is given. Sect. III recalls the EKF fundamentals, and the innovative application to a synchronous rotating reference frame. Sect. IV presents the hardware structure. Experimental results, obtained on a laboratory prototype, are reported in Sect.V. Conclusive remarks close the paper.

II. IPM SYNCHRONOUS MOTOR

Interior permanent magnet synchronous motors give excellent performance in terms of torque-to-ampere and torque-to-volume ratios, minimizing both the inverter cost and the drive dimensions. The simple construction and mechanical robustness make them ideal for high-speed applications. The most suitable motor structure considers an anisotropic rotor, with a fairly reduced permanent magnet flux. Fig.1 shows two pole-pitches of the motor used in the experimental setup. A commercial 24-slot induction motor stator has been used, with a single-layer winding.

The financial support of Elite Research Program by Texas Instruments Company is gratefully acknowledged.

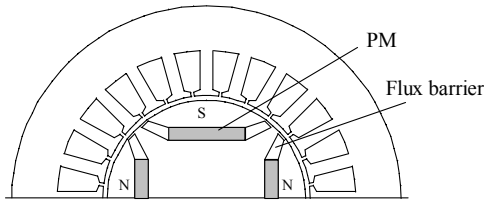


Fig.1 – Cross-section of IPM motor

Motor design has been carried out by a finite elements analysis (FEA) tool, which has also provided a first rough parameters identification. They have been validated by a batch of measurements, carried out on the prototype. The motor data are reported in App. A.

Let us place the direct (d) axis on PM flux direction, while quadrature (q) axis leads d axis by $\pi/2$ electrical degrees. Fig.2 reports the flux paths, produced by the permanent magnet (PM) only and by q -axis current respectively.

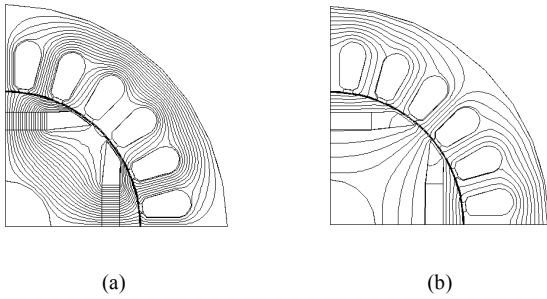


Fig.2 – Flux paths: a) PM only, b) q -axis current flux.

As said before, buried magnets yield different paths for d and q axes. In particular, quadrature path runs mostly in the stator and rotor iron, and thus the quadrature inductance L_q is higher than L_d , and it is also more subjected to saturation than L_d . Direct and quadrature flux linkages as function of currents, for the IPM motor of Fig.2 are reported in Fig.3.

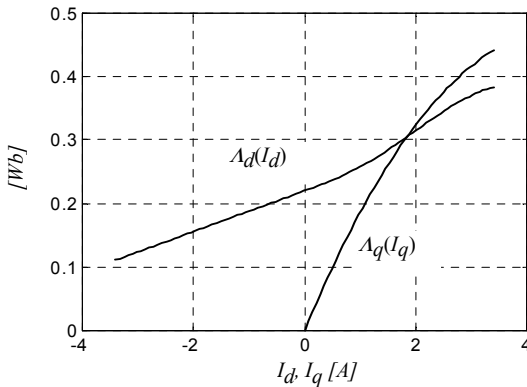


Fig.3 - Two axes flux linkages as functions of the two axes currents

The curves represented in Fig.3 put in evidence that L_d can be roughly kept constant in the motor model, while L_q saturation cannot be neglected at all. Conversely, tests with different

levels of i_d and i_q currents have revealed a fairly low magnetic cross coupling between the d and q axes, which will be neglected in the following. The effect of this simplification will be ascribed to the model imperfection.

III. EKF DESIGN CONSIDERATIONS

Motor model can be derived both in a stationary orthogonal reference frame α, β , fixed to the stator, or in a synchronous reference frame d, q , fixed to the rotor flux. Usually, the EKF algorithm for PMSM is derived for the former representation [12-14]. In the following, some considerations on this crucial choice are developed. Advantages and drawbacks of each solution are depicted.

For both models, the so-called infinite inertia hypothesis will be assumed. This is the same as neglecting the influence of speed variations within the time step T_c in the model itself. The robustness of the EKF convergence to approximate load equations has been already documented in [14]. When not otherwise specified, inductances will be considered independent from currents, i.e. saturation effects will be neglected. Actually, a precise mathematical model should consider either apparent (Λ/I) and differential ($d\Lambda/dI$) inductances, that could be both derived from curves of Fig.3. The complexity would be unbearable. It has been chosen to ascribe this imperfection to the model and exploit the EKF robustness to get a good state estimation, in spite of the model approximations.

A. IPM Motor Stationary Model.

The two axis currents (i_α, i_β), the rotor electrical position ϑ_{me} and speed ω_{me} can be selected as system state variables. With the above assumptions, the two IPM motor voltage equations and the two load dynamic equations can be rearranged in a canonical state-space form as follows:

$$\begin{cases} \frac{di_\alpha}{dt} = \frac{1}{L_d L_q} [D_1(\vartheta_{me})(u_\alpha - R i_\alpha - e_\alpha) - \omega_{me} B_1(\vartheta_{me}) i_\alpha + \\ \quad - \omega_{me} C_1(\vartheta_{me}) i_\beta - L_1(\vartheta_{me})(u_\beta - R i_\beta - e_\beta)] \\ \frac{di_\beta}{dt} = \frac{1}{L_d L_q} [D_2(\vartheta_{me})(u_\beta - R i_\beta - e_\beta) - \omega_{me} B_2(\vartheta_{me}) i_\alpha + \\ \quad - \omega_{me} C_2(\vartheta_{me}) i_\beta - L_1(\vartheta_{me})(u_\alpha - R i_\alpha - e_\alpha)] \\ \frac{d\omega_{me}}{dt} = 0 \\ \frac{d\vartheta_{me}}{dt} = \omega_{me} \end{cases} \quad (1)$$

where:

$$L_1(\vartheta_{me}) = \frac{L_d - L_q}{2} \sin(2\vartheta_{me}) \quad (2)$$

$$L_2(\vartheta_{me}) = \frac{L_d - L_q}{2} \cos(2\vartheta_{me}) \quad (3)$$

$$B_1(\vartheta_{me}) = -L_1(\vartheta_{me})(L_d - L_q) \quad (4)$$

$$B_2(\vartheta_{me}) = L_2(\vartheta_{me})(L_d + L_q) + \frac{(L_d - L_q)^2}{2} \quad (5)$$

$$C_1(\vartheta_{me}) = L_2(\vartheta_{me})(L_d + L_q) - \frac{(L_d - L_q)^2}{2} \quad (6)$$

$$C_2(\vartheta_{me}) = L_1(\vartheta_{me})(L_d + L_q) \quad (7)$$

$$D_1(\vartheta_{me}) = \frac{L_d + L_q}{2} - L_2(\vartheta_{me}) \quad (8)$$

$$D_2(\vartheta_{me}) = \frac{L_d + L_q}{2} + L_2(\vartheta_{me}) \quad (9)$$

It is worth to recall that in practice inductance L_q is sensible to saturation (Fig.3) and therefore it would be in general a function of i_α , i_β and ϑ_{me} , with further model complication. The phase voltages u_α and u_β represent the system input, while i_α and i_β can be considered as system output. The influence of rotor position on the stator inductances complicates the Jacobian matrix of the system, that is an essential element in the prediction step of the EKF algorithm. The CPU time required for the four-by-four Jacobian matrix calculation could overwhelm the DSP time-sharing routine management. As a spot, the explicit expression of the Jacobian element $F_{1,3} = d(di_\alpha/dt)/d\omega_{me}$ is given hereafter.

$$F_{1,3} = \frac{1}{L_d L_q} [-B_1 i_\alpha - C_1 i_\beta + D_1 A_{mg} \sin(\vartheta_{me}) + L_1 A_{mg} \cos(\vartheta_{me})] \quad (10)$$

and symbols are defined by (2)-(9). It can be proved that no modification occurs to the innovation step respect to the isotropic case, described for example in [14]. As a design hint, it has been found that the amount of floating point operations (FLOPS) in the EKF prediction step for IPM motor is almost triple the case of the isotropic motor.

B. IPM Motor Synchronous Model.

In order to obviate the large computing time requested by the EKF algorithm in the stationary reference frame, and to simplify the formal expression of the equations, the impact of a coordinates transformation has been evaluated. Then, the whole EKF algorithm has been re-written in the synchronous frame and the new state vector becomes $\mathbf{x} = [i_d \ i_q \ \omega_{me} \ \vartheta_{me}]'$. The system state equations are the following:

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}(t))\mathbf{x}(t) + \mathbf{G} \mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{H} \mathbf{x}(t) \end{aligned} \quad (11)$$

where $\mathbf{u} = [u_d \ u_q]'$ and $\mathbf{y} = [i_d \ i_q]'$ are the input and output vectors respectively. The system state matrices are defined as:

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (12)$$

$$\mathbf{f}(\mathbf{x}(t)) = \begin{bmatrix} -\frac{R}{L_d} & \omega_{me} \frac{L_q}{L_d} & 0 & 0 \\ -\omega_{me} \frac{L_d}{L_q} & -\frac{R}{L_q} & -\frac{A_{mg}}{L_q} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (13)$$

$$\mathbf{G} = \begin{bmatrix} \frac{1}{L_d} & 0 & 0 & 0 \\ 0 & \frac{1}{L_q} & 0 & 0 \end{bmatrix} \quad (14)$$

Although a remarkable change in algorithm syntax, equations appear greatly simpler than those expressed by (1). The choice of the input vector $\mathbf{u}$ is worth a deeper insight. In the proposed implementation, the measurement of the phase voltages is replaced by the reference space vector modulation (SVM) signals. The actual dq voltage reference delivered by current control may be altered by SVM saturation and dead time routines. Therefore, the use of corrected quantities in the dq frame implies an additional inverse $\alpha\beta/dq$ transformation, operated on voltage references directly derived from definitive SVM timings.

C. Extended Kalman Filter Equations.

The extended Kalman filter is an optimal estimator in the least-square sense for estimating the states of dynamic non-linear systems. It consists of two main parts, namely the *prediction step* and the *innovation step*. The former performs a state estimation based on the best available motor model, while the latter filters and corrects the predicted state by measuring the actual motor phase currents.

In this work, the EKF code originally developed for a sensorless SPM motor [14] has been modified to extend the EKF to the IPM motor. For a full-digital implementation, the system model (11) has been discretised every kT_c , by a rectangular approximation:

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{F}_d(\mathbf{x}_k)\mathbf{x}_k + \mathbf{G} \mathbf{u}_k + \mathbf{v}_k \\ \mathbf{y}_k &= \mathbf{H} \mathbf{x}_k + \boldsymbol{\mu}_k \end{aligned} \quad (15)$$

According to the EKF theory, the model (15) also includes a statistical description of inaccuracies. As usual, in absence of more precise system knowledge, both model and measurement disturbances have been described by the zero-mean Gaussian random vectors $\mathbf{v}_k = \mathbf{v}(kT_c)$ and $\boldsymbol{\mu}_k = \boldsymbol{\mu}(kT_c)$, whose variance

matrices are $\mathbf{Q}$ and $\mathbf{R}$ respectively. The discretised system matrix $\mathbf{F}_d$ is derived from (13) as follows:

$$\mathbf{F}_d(\mathbf{x}_k) = 1 + T_c \mathbf{f}(\mathbf{x}(kT_c)) \quad (16)$$

At time $t_k = kT_c$ the optimal state estimate $\mathbf{x}_{e|k}$ and the estimation error covariance matrix $\mathbf{P}_{k|k}$ are obtained through a simplified version of the EKF algorithm [15], summarised in Tab.I.

Tab.I - EKF algorithm

Prediction step
$\mathbf{x}_{e k-1} = \mathbf{F}_d(\mathbf{x}_{e k-1})\mathbf{x}_{e k-1} + \mathbf{G}\langle \mathbf{u}_{k-1} \rangle$
$\mathbf{P}_{k k-1} = \mathbf{P}_{k-1 k-1} + (\mathbf{F}_{k-1}\mathbf{P}_{k-1 k-1} + \mathbf{P}_{k-1 k-1}\mathbf{F}_{k-1}')T_c + \mathbf{Q}$
Innovation step
$\mathbf{x}_{e k} = \mathbf{x}_{e k-1} + \mathbf{K}_k [\mathbf{y}_k - \mathbf{H}\mathbf{x}_{e k-1}]$
$\mathbf{P}_{k k} = \mathbf{P}_{k k-1} - \mathbf{K}_k \mathbf{H} \mathbf{P}_{k k-1}$
Kalman gain
$\mathbf{K}_k = \mathbf{P}_{k k-1} \mathbf{H}' (\mathbf{H} \mathbf{P}_{k k-1} \mathbf{H}' + \mathbf{R})^{-1}$

The prediction of the state covariance requires the on-line computation of the Jacobian matrix $\mathbf{F}$, defined as

$$\mathbf{F}_{k-1} = \left. \frac{\partial \mathbf{f}(\mathbf{x}(t)) \mathbf{x}(t)}{\partial \mathbf{x}} \right|_{\mathbf{x}=\mathbf{x}_{e|k-1}} \quad (17)$$

For the sake of completeness, the explicit calculation of (17) is drawn in App. B. The main disadvantage of the synchronous model resides in the innovation step of the EKF algorithm, where predicted and actual currents are compared. Opposite to the stationary frame case, the measured quantities need to undergo a coordinate transformation, in which it is involved the estimate electrical position itself. In presence of uncertainties in the model, this may yield a constant error in the position estimate. It has been found that the EKF algorithm shows the greatest sensibility to L_q mismatch.

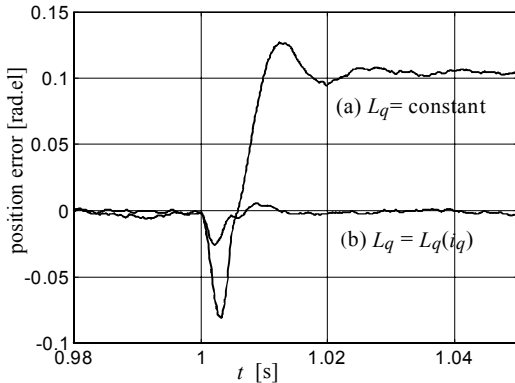


Fig.4 – (a) Position error due to L_q saturation. (b) LUT compensation

By an off-line comparison between estimated and measured position, for different levels of the i_q current, such sensibility has been exploited to get a fine tuning of the relation $L_q(i_q)$,

that is used instead of a constant L_q value in (15). Fig. 4 shows the steady state error between the estimated and actual position ($\vartheta_{me} - \vartheta_{e|me}$), as function of time, at no load and after a step of 50% of rated torque. Case (a) refers to a constant L_q in the model and (b) uses on-line $L_q(i_q)$ retrieval from a memory look-up table (LUT). Even if (15) remains a simplified model, the use of $L_q(i_q)$ yields an effective abatement of the steady state position error estimate.

IV. DRIVE STRUCTURE

The experimental IPM motor drive designed for this work featured a wide flux-weakening region. As a counterpart, the presence of both reluctance and PM torque requires involved control strategies, to guarantee effective drive operations at all conditions. The block schematic of the complete drive structure is reported in Fig.5.

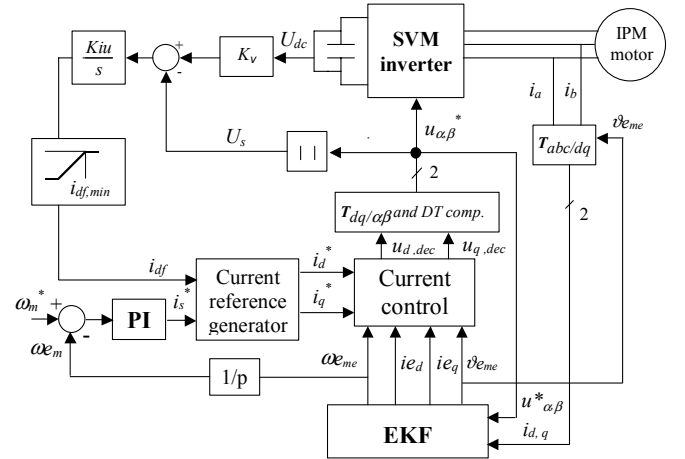


Fig.5 - IPM motor sensorless drive block schematic.

The specific target of this work was the test of EKF behaviour at high speed. A smooth transition from the constant torque to the flux-weakening region is guaranteed by an external voltage closed loop regulation, as depicted in Fig.5. The phase voltage amplitude is compared with a limit, proportional to the dc link voltage. As mentioned before, accurate deadtime compensation allows the use of reference voltages instead of the actual ones. When the limit is exceeded, the error is integrated and a proper flux weakening action is triggered, through the delivery of an additional direct current component i_{df} (Fig.5). The choice of the voltage limit K_v has to be harmonised with that of the integral gain K_{iu} , that in turn must preserve the overall system stability. It is worth to note that the greater K_v , the closer the voltage limit to the inverter saturation. If this is the case, a prompt flux weakening action is required. Of course, this calls for higher values of K_{iu} , whose upper limit is yet imposed by system stability. In this work, it was set $K_v=0.51$. As a simple rule of thumb, it is reasonable to choose K_{iu} so that in presence of the maximum

input error the maximum flux-weakening component I_N is delivered within a fixed time. This time is preferably much larger (20-30 times) than the i_d rise time t_{ri} , to maintain a good separation between the voltage and the current control loops dynamic. Since the amplitude of the maximum undistorted voltage (before entering the hexagonal saturation) for a SVM inverter is $U_{dc} \sqrt{3}$, the integral gain can be calculated as

$$K_{iu} = \frac{I_N}{30t_{ri}(\sqrt{3} - K_v)U_{dc}} \quad (18)$$

Several simulation results have confirmed that if K_{iu} is chosen sensibly smaller than (18) the voltage loop is no more able to follow the transient voltage produced by the current control, and inverter saturation occurs. Conversely, if the voltage loop is too fast, instability due to system delays has been experienced.

Another crucial parameter is the maximum negative current $i_{df,min}$, which limits the output of the integrator (Fig.5). It is worth to perform an adaptive limitation, according to the actual working point, by setting

$$i_{df,min} = I_N - i_{d,n}^* \quad (19)$$

where I_N is the rated stator current and $i_{d,n}^*$ is the direct current that the current reference generator would give in absence of voltage limitation. Several experimental tests have highlighted that if a fixed limitation would be chosen instead, instability may arise at load detaching during operation far above the base speed [4].

A. The Current Reference Generator.

In the drive schematic of Fig.5, the output of the proportional-integral (PI) speed regulator is the (signed) amplitude of the stator current space vector i_s^* . The well-known maximum torque per ampere strategy has been used to link the PI output to the torque production. For a given i_d, i_q pair and under the hypothesis of system linearity the electromagnetic torque is given by

$$\tau = \frac{3}{2} p i_q [A_{mg} - (L_q - L_d) i_d] \quad (20)$$

The direct current reference i_d^* is expressed as function of i_q^* and the current reference amplitude i_s^*

$$i_d^* = -\sqrt{(i_s^*)^2 - (i_q^*)^2} \quad (21)$$

It is possible to determine the value of the reference i_q^* that maximises the torque, by imposing

$$\frac{\partial \tau}{\partial i_q^*} = 0 \quad (22)$$

that yields to

$$i_q^* = \text{sign}(\tau) \sqrt{\frac{(2i_s^*)^2 - i_{da}^2 + \sqrt{i_{da}^4 + 8(i_{da} i_s^*)^2}}{8}} \quad (23)$$

where $i_{da} = A_{mg}/(L_d - L_q)$. It is soon evident that, in presence of deep saturation of the q -path, (21) and (23) are only rough and ready approximations of the correct d - q current references. They have been taken as starting point for the on the field tuning of the drive. Into details, the IPM motor drive has been operated in current mode in a test bench capable of torque measurement. For each reference value of the stator current, it has been selected the couple i_d, i_q which yields the maximum motor torque production. Fig.6 shows the resulting d - q axes current references as function of stator current amplitude.

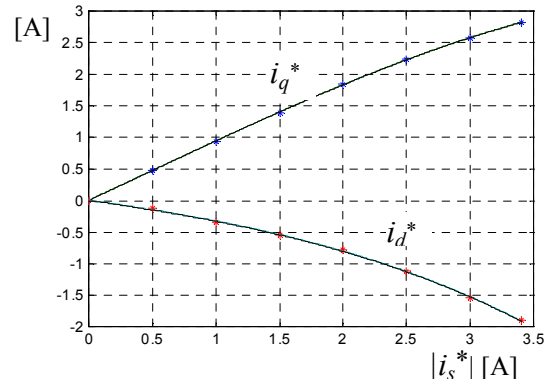


Fig.6 – Current reference generator in-out curves

The i_d, i_q references of Fig.6 have been stored in a look-up table, which has been used as optimal current reference generator in the constant torque region.

B. The Current Control.

A proper current control can be achieved in the synchronous d - q coordinates system by conventional PI controllers. In this work, PI parameters have been tuned to get a bandwidth of approximately 600 Hz, with simultaneous two-phases current sampling at 7.2 kHz, which corresponds to $T_c = 140 \mu s$. Sampling is triggered at the beginning of each SVM switching period, so that the measured signals are less affected by the ripple due to voltage modulation.

Particular care has been paid for the management of current regulators output and integral limitation. Output limitations have been let to SVM algorithm, which inherently constrains the produced voltage vector within a hexagonal boundary.

Anti-wind up management is obtained by comparing the actual voltage references with the input errors. In presence of saturation, the control freezes the integral action whenever the voltage reference and the input error have the same sign [4].

In the synchronous frame, the motor model presents coupling terms between d and q axes. These terms can be considered as disturbances that complicate the design of PI controllers and reduce the overall dynamic performances. A decoupling and feedforward algorithm has been implemented in the drive, as sketched in Fig.7.

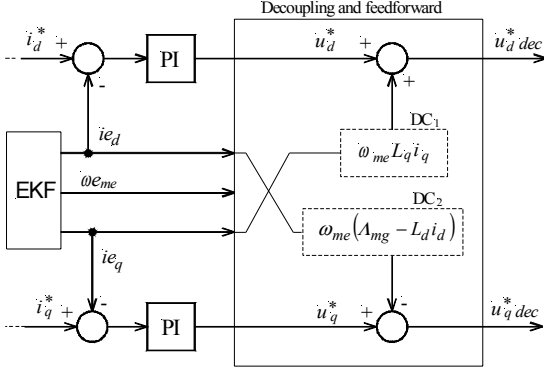


Fig.7 – Decoupling scheme in current control

A precise decoupling action (blocks DC_1 and DC_2) has been obtained by including the L_q inductance saturation.

V. EXPERIMENTAL RESULTS

The EKF algorithm and the speed and current control loops, including the dead time compensation, have been implemented on a TMS320C31 Floating Point DSP, with 33.3 ns instruction cycle. All software procedures are written in C language, optimised for shortest execution time. Digital I/O management and switching pattern generation are handled by a slave Fixed Point TMS320P14 DSP, with 160 ns instruction cycle, featuring six PWM channels, with 80 ns period resolution and 16 individual bit-selectable I/O. Control software routines for the slave processor have been written in Assembly language, for a closer link with the embedded hardware peripherals. Phase currents have been measured by two parallel auto calibrating 16-bit 10 μ s A/D converters, while speed reference and DC link bus voltage are measured by two 12-bit 3 μ s A/D converters. From the software side, the first step was to tune accurately the current and speed PI regulators, by using a resolver as a position sensor. While maintaining the drive closed on measured quantities, a first batch of experiments was devoted to find out the range of values of Q matrix that shorten the EKF convergence. A fine-tuning of Q and R matrices was then obtained by minimising the settling time of the speed control loop, with feedback closed on estimated quantities [15].

When implemented in the stationary reference frame, the EKF may exhibit a wrong convergence to the solution ($-\omega_r \vartheta + \pi$). As found in [14], the wrong solution is maintained by the innovation step that continuously updates the estimate, compensating for the gap between the actual and predicted currents and voltages. A side advantage of the EKF

implementation in the d - q reference is that the wrong solution, also known as "slip condition", does not fit the voltage equations of the model, thus preserving the system from a misleading convergence.

The correct operation of the sensorless drive in the flux-weakening region is documented in Fig.8. It reports a no-load acceleration ramp, from 0.15 to 2.2 times the base speed Ω_B .

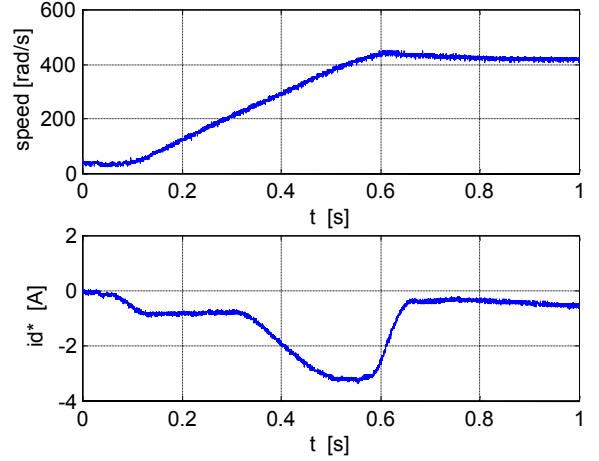


Fig.8 – Ramp acceleration to 2.2 Ω_B and i_d^* generation

The same figure shows also the i_d current component generated by the current reference generator during the speed ramp. It is easy to recognise that the delivery of the flux weakening component i_{df} is triggered around the base speed Ω_B . At the end of the speed ramp the cancellation of the inertial torque acts as a load detaching. The amplitude of the current vector suddenly reduces, and less flux weakening component is required, as evident in the lower part of Fig.8.

As mentioned earlier, the voltage loop is committed to avoid SVM saturation. Fig.9 reports the phase voltage amplitude during an acceleration from zero to 2.2 Ω_B .

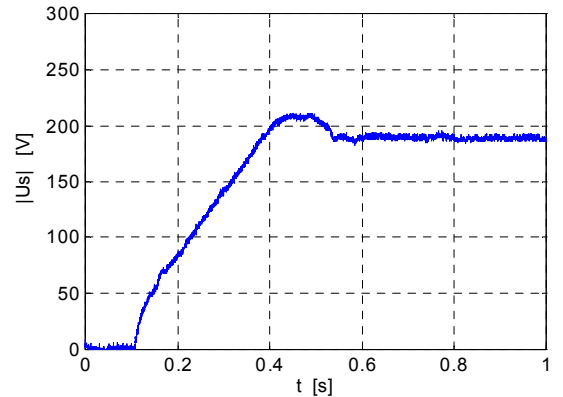


Fig.9 - Phase voltage amplitude during ramp acceleration.

The steady state fixed voltage limitation was moving along smoothly, without harmful effects on current control.

The last batch of measurements was carried out to investigate the drive response to different load conditions, in the flux-weakening region. Fig.10 shows the speed and i_q current

behaviour to a positive load step variation of 30% T_B , at 1.65 Ω_B .

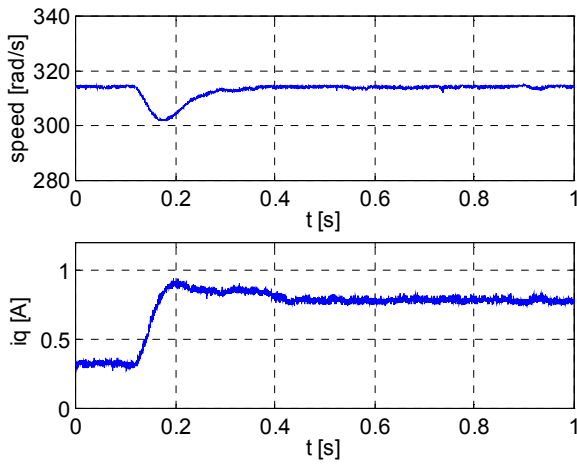


Fig.10 – Response to a load detach in flux weakening.

The transient is well recovered and the EKF maintains the convergence, demonstrating excellent properties also in this particular application.

VI. CONCLUSIONS

This paper has presented the application of an extended Kalman filter to the estimation of speed and position in an IPM sensorless drive. The research was focused on high-speed applications, above the motor base speed. The assumption was that at high speed the major – and still unsolved – drawbacks of EKF were of minor relevance, while the EKF robustness could profitably be used to get a reliable sensorless drive. The work implied an extensive theoretical approach, with a substantial differentiation from those existing in literature. A complete prototype was realised in the laboratory, on which all the algorithms were implemented and tested. The paper reports the results, which prove the effectiveness of the study. The industrialisation of the prototype with a business partner is already under consideration.

LIST OF SYMBOLS

x	System state vector
$\hat{x}$	Estimated state vector
u	System input vector
y	System output vector
F	Jacobian of system state functional
H	System output matrix
K	Kalman gain matrix
P	State error covariance matrix
Q	Model noise covariance matrix
R	Measurement noise covariance matrix
i_s	Stator current vector
R_s	Stator resistance
$L_{d,q}$	Stator direct, quadrature inductance
Λ_{mg}	PM flux linkage

T_c	Sampling period
p	Motor pole pairs
$\omega_{me}, \vartheta_{me}$	Electrical rotor speed, position
$\hat{\omega}_{me}, \hat{\vartheta}_{me}$	Estimated electrical speed, position
ω_m	Mechanical speed
Subscripts, superscripts	
d, q	Synchronous frame quantities
α, β	Stationary frame quantities
k	Sampling index
$k k-1$	Predicted estimate
$k k$	Optimal estimate
0	Initial value
$'$	Transposed matrix
$*$	Reference quantity
B	Base quantity
u	Unsaturated parameter

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APPENDIX A – MOTOR DATA AND PARAMETERS

The IPM motor prototype features neodymium-iron-boron (NeFeB) rare-earth magnets. The main parameters are reported in Tab.II.

Tab. II - IPM Motor Data

Quantity	Value	Unit
Base torque T_B	3.3	Nm
Base speed Ω_B	190	rad /s
Max phase voltage U_B	192	V _{peak}
Nominal current I_{nom}	2.4	A _{rms}
Poles pairs, p	2	-
R_s (@ 50°C)	5	Ω
L_d	0.032	H
L_{qu} (unsaturated)	0.2	H
L_q ($I_q=I_{nom}$)	0.16	H
PM Flux linkage Λ_{mg}	0.215	Vs/rad

The DC link was fixed to 370 V, monitored on-line to keep constant the SVM inverter gain.

APPENDIX B – CALCULATION OF THE JACOBIAN MATRIX F

At time $t_{k-I}=(k-1)T_c$ the Jacobian matrix F is defined as in (17), here reported for ease of reference:

$$F_{k-1} = \left. \frac{\partial f(x(t))x(t)}{\partial x} \right|_{x=xe_{k|k-1}} = \left. \frac{\partial \dot{x}}{\partial x} \right|_{x=xe_{k|k-1}}$$

From (13) and (14) it holds

$$\dot{x} = \begin{bmatrix} -\frac{R}{L_d}i_d + \frac{\omega_{me}L_q}{L_d}i_q + \frac{u_d}{L_d} \\ -\frac{\omega_{me}(L_d i_d + \Lambda_{mg})}{L_q} - \frac{Ri_q}{L_q} + \frac{u_q}{L_q} \\ 0 \\ \omega_{me} \end{bmatrix}$$

F is therefore a four-by-four matrix,

$$F = \begin{bmatrix} F_{11} & F_{12} & F_{13} & 0 \\ F_{21} & F_{22} & F_{23} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & F_{43} & 0 \end{bmatrix}$$

whose elements at time $t_{k-I}=(k-1)T_c$ can be derived as follows:

$$F_{11} = \left. \frac{\partial}{\partial i_d} \left(\frac{di_d}{dt} \right) \right|_{x=xe_{k|k-1}} = -\frac{R}{L_d}$$

$$F_{12} = \left. \frac{\partial}{\partial i_q} \left(\frac{di_d}{dt} \right) \right|_{x=xe_{k|k-1}} = \frac{L_q}{L_d} \omega_{me,k|k-1}$$

$$F_{13} = \left. \frac{\partial}{\partial \omega_{me}} \left(\frac{di_d}{dt} \right) \right|_{x=xe_{k|k-1}} = \frac{L_q}{L_d} i_{e_{q,k|k-1}}$$

$$F_{21} = \left. \frac{\partial}{\partial i_q} \left(\frac{di_q}{dt} \right) \right|_{x=xe_{k|k-1}} = -\frac{L_d}{L_q} \omega_{me,k|k-1}$$

$$F_{22} = \left. \frac{\partial}{\partial i_q} \left(\frac{di_q}{dt} \right) \right|_{x=xe_{k|k-1}} = -\frac{R}{L_q}$$

$$F_{23} = \left. \frac{\partial}{\partial \omega_{me}} \left(\frac{di_q}{dt} \right) \right|_{x=xe_{k|k-1}} = -\frac{L_d i_{e_{d,k|k-1}} + \Lambda_{mg}}{L_q}$$

$$F_{43} = \left. \frac{\partial}{\partial \omega_{me}} \left(\frac{d\vartheta_{me}}{dt} \right) \right|_{x=xe_{k|k-1}} = 1$$